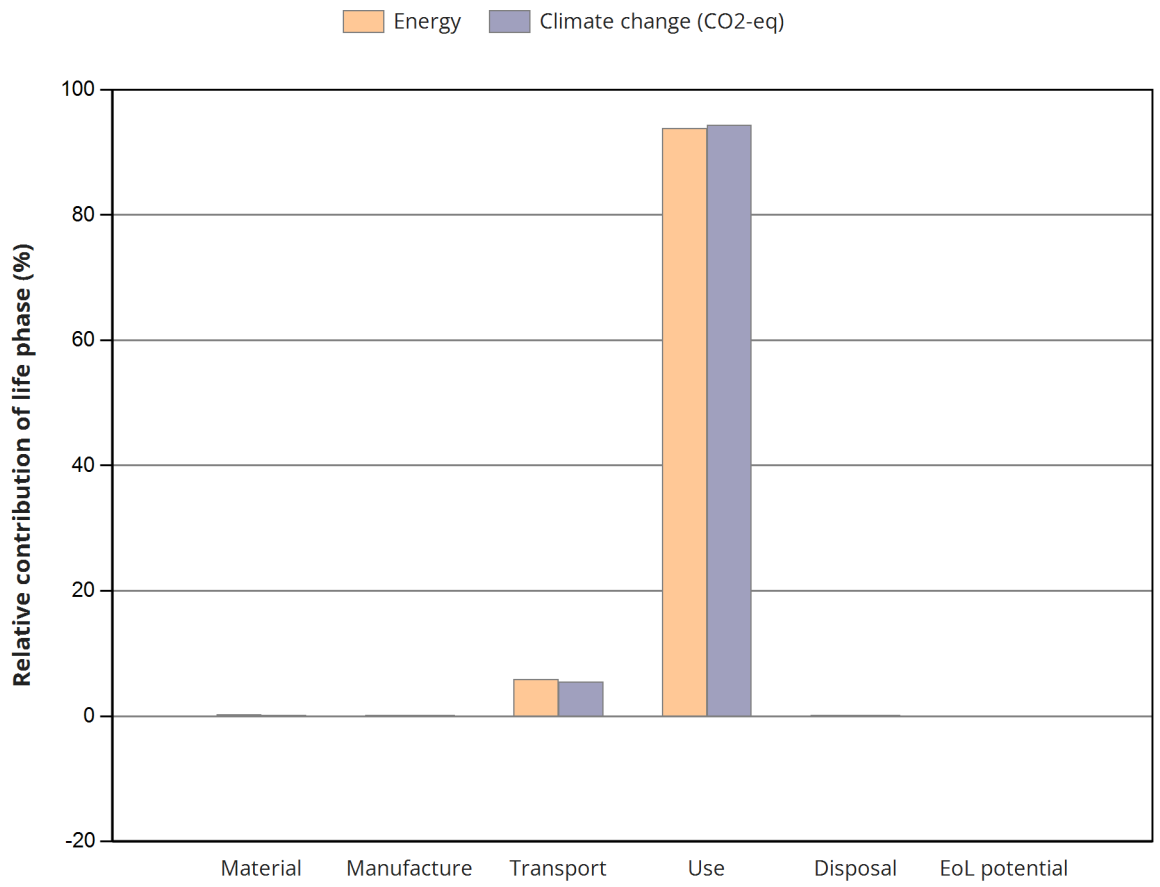
Eco Audit Report

Product name	Wind Blade
Country of use	The Whole World
Product life (years)	3

Summary:



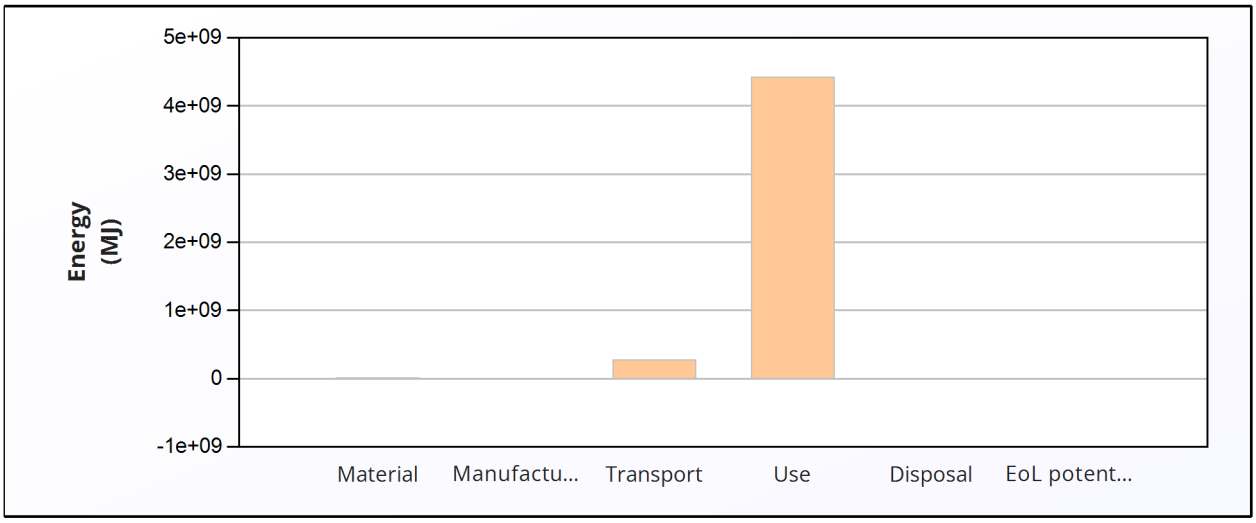
Energy details Climate change (CO2-eq) details

Phase	Energy (MJ)	Energy (%)	Climate change (CO2-eq) (kg)	Climate change (CO2-eq) (%)
Material	1.07e+07	0.2	6.43e+05	0.2
Manufacture	7.32e+05	0.0	5.48e+04	0.0
Transport	2.8e+08	5.9	1.85e+07	5.5
Use	4.42e+09	93.8	3.18e+08	94.3
Disposal	2.49e+04	0.0	1.75e+03	0.0
Total (for first life)	**4.71e+09**	**100**	**3.38e+08**	**100**
End of life potential	-3.27e+06		-1.57e+05	

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ENERGY ANALYSIS

Summary



	Energy (MJ/year)
Equivalent annual environmental burden (averaged over 3 year product life):	1.57e+09

DETAILED BREAKDOWN OF INDIVIDUAL LIFE PHASES

MATERIAL:

Summary

Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass (kg)	Energy (MJ)	%
Mid Carbon Steel Blade	Medium carbon steel	Virgin (0%)	23	1000	2.3e+04	7.2e+05	6.8
Stainless Steel Blade	Stainless steel	Virgin (0%)	23	1000	2.3e+04	1.6e+06	14.5
Super Ni Blade	Nickel-based superalloys	Virgin (0%)	23	1000	2.3e+04	7.7e+06	72.0
Low key Blade	Low alloy steel	Virgin (0%)	23	1000	2.3e+04	7.2e+05	6.8
Total				**4000**	**9.1e+04**	**1.1e+07**	**100**

*Typical: Includes 'recycle fraction in current supply'

***User-defined material

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MANUFACTURE:

Summary

Component	Process	Amount processed	Energy (MJ)	%
Mid Carbon Steel Blade	Roll forming	2.3e+04 kg	1.2e+05	16.1
Stainless Steel Blade	Roll forming	2.3e+04 kg	1.2e+05	15.9
Super Ni Blade	Casting	2.3e+04 kg	2.4e+05	33.4
Low key Blade	Casting	2.3e+04 kg	2.5e+05	34.5
Total			**7.3e+05**	**100**

TRANSPORT:

Summary

BREAKDOWN BY TRANSPORT STAGE

Stage name	Transport type	Distance (km)	Energy (MJ)	%
Ground unit	Truck 16-32t, EURO 6	1e+06	2.7e+08	97.7
Sea unit	Sea, container ship	5.3e+05	6.5e+06	2.3
Total		**1.6e+06**	**2.8e+08**	**100**

BREAKDOWN BY COMPONENTS

Component	Mass (kg)	Energy (MJ)	%
Mid Carbon Steel Blade	2.3e+04	7e+07	25.0
Stainless Steel Blade	2.3e+04	7e+07	25.0
Super Ni Blade	2.3e+04	7e+07	25.0
Low key Blade	2.3e+04	7e+07	25.0
Total	**9.1e+04**	**2.8e+08**	**100**

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Summary

USE:

MOBILE MODE

Fuel and mobility type	Kerosene - long haul aircraft
Country of use	The Whole World
Product mass (kg)	9.1e+04
Distance (km per day)	1e+04
Usage (days per year)	2.5e+02
Product life (years)	3

RELATIVE CONTRIBUTION OF STATIC AND MOBILE MODES

Mode	**Energy (MJ)**	**%**
Static	0	
Mobile	4.4e+09	100.0
Total	**4.4e+09**	**100**

BREAKDOWN OF MOBILE MODE BY COMPONENTS

Component	**Energy (MJ)**	**%**
Mid Carbon Steel Blade	1.1e+09	25.0
Stainless Steel Blade	1.1e+09	25.0
Super Ni Blade	1.1e+09	25.0
Low key Blade	1.1e+09	25.0
Total	**4.4e+09**	**100**

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Summary

Disposal:

Component	End of life option	Energy (MJ)	%
Mid Carbon Steel Blade	Landfill	4.5e+03	18.2
Stainless Steel Blade	Landfill	4.5e+03	18.2
Super Ni Blade	Downcycle	1.1e+04	45.5
Low key Blade	Landfill	4.5e+03	18.2
Total		**2.5e+04**	**100**

EoL potential:

Component	End of life option	Energy (MJ)	%
Mid Carbon Steel Blade	Landfill	0	0.0
Stainless Steel Blade	Landfill	0	0.0
Super Ni Blade	Downcycle	-3.3e+06	100.0
Low key Blade	Landfill	0	0.0
Total		**-3.3e+06**	**100**

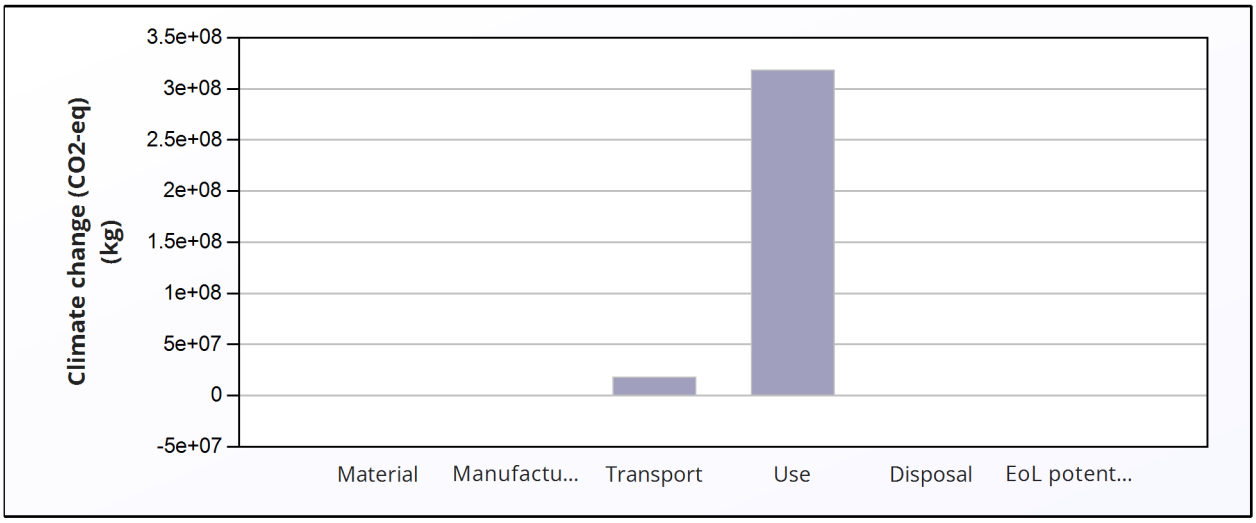
Summary

Notes:

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CLIMATE CHANGE (CO2-EQ) ANALYSIS

Summary



	CO2 (kg/year)
Equivalent annual environmental burden (averaged over 3 year product life):	1.13e+08

DETAILED BREAKDOWN OF INDIVIDUAL LIFE PHASES

MATERIAL:

Summary

Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass (kg)	Climate change (CO2-eq) (kg)	%
Mid Carbon Steel Blade	Medium carbon steel	Virgin (0%)	23	1000	2.3e+04	5.3e+04	8.2
Stainless Steel Blade	Stainless steel	Virgin (0%)	23	1000	2.3e+04	1.3e+05	20.7
Super Ni Blade	Nickel-based superalloys	Virgin (0%)	23	1000	2.3e+04	4e+05	62.9
Low key Blade	Low alloy steel	Virgin (0%)	23	1000	2.3e+04	5.3e+04	8.2
Total				**4000**	**9.1e+04**	**6.4e+05**	**100**

*Typical: Includes 'recycle fraction in current supply'

***User-defined material

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Manufacture:

Summary

Component	Process	Amount processed	Climate change (CO2-eq) (kg)	%
Mid Carbon Steel Blade	Roll forming	2.3e+04 kg	8.9e+03	16.2
Stainless Steel Blade	Roll forming	2.3e+04 kg	8.7e+03	15.9
Super Ni Blade	Casting	2.3e+04 kg	1.8e+04	33.4
Low key Blade	Casting	2.3e+04 kg	1.9e+04	34.5
Total			**5.5e+04**	**100**

Transport:

Summary

Breakdown by transport stage

Stage name	Transport type	Distance (km)	Climate change (CO2-eq) (kg)	%
Ground unit	Truck 16-32t, EURO 6	1e+06	1.8e+07	97.3
Sea unit	Sea, container ship	5.3e+05	4.9e+05	2.7
Total		**1.6e+06**	**1.9e+07**	**100**

Breakdown by components

Component	Mass (kg)	Climate change (CO2-eq) (kg)	%
Mid Carbon Steel Blade	2.3e+04	4.6e+06	25.0
Stainless Steel Blade	2.3e+04	4.6e+06	25.0
Super Ni Blade	2.3e+04	4.6e+06	25.0
Low key Blade	2.3e+04	4.6e+06	25.0
Total	**9.1e+04**	**1.9e+07**	**100**

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Summary

USE:

MOBILE MODE

Fuel and mobility type	Kerosene - long haul aircraft
Country of use	The Whole World
Product mass (kg)	9.1e+04
Distance (km per day)	1e+04
Usage (days per year)	2.5e+02
Product life (years)	3

RELATIVE CONTRIBUTION OF STATIC AND MOBILE MODES

Mode	**Climate change (CO2-eq) (kg)**	**%**
Static	0	
Mobile	3.2e+08	100.0
Total	**3.2e+08**	**100**

BREAKDOWN OF MOBILE MODE BY COMPONENTS

Component	**Climate change (CO2-eq) (kg)**	**%**
Mid Carbon Steel Blade	8e+07	25.0
Stainless Steel Blade	8e+07	25.0
Super Ni Blade	8e+07	25.0
Low key Blade	8e+07	25.0
Total	**3.2e+08**	**100**

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Summary

DISPOSAL:

Component	End of life option	Climate change (CO2-eq) (kg)	%
Mid Carbon Steel Blade	Landfill	3.2e+02	18.2
Stainless Steel Blade	Landfill	3.2e+02	18.2
Super Ni Blade	Downcycle	7.9e+02	45.5
Low key Blade	Landfill	3.2e+02	18.2
Total		**1.7e+03**	**100**

EOL POTENTIAL:

Component	End of life option	Climate change (CO2-eq) (kg)	%
Mid Carbon Steel Blade	Landfill	0	0.0
Stainless Steel Blade	Landfill	0	0.0
Super Ni Blade	Downcycle	-1.6e+05	100.0
Low key Blade	Landfill	0	0.0
Total		**-1.6e+05**	**100**

Summary

NOTES:

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